

Vulnerability Assessment Report

1. Title Page

Project: Vulnerability Assessment of a Live Website

Tool Used: OWASP ZAP (Zed Attack Proxy)

Target Website: <http://demo.testfire.net>

Prepared By: Bulla Jaswanth

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2. Introduction

This report presents the results of a vulnerability assessment conducted on a web application using OWASP ZAP. The purpose of this assessment is to identify security vulnerabilities, evaluate their risk levels, and recommend appropriate mitigation strategies.

3. Objectives

- Identify common web vulnerabilities
 - Classify risks (High, Medium, Low)
 - Provide clear explanations of issues
 - Suggest remediation steps
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4. Methodology

The following steps were followed during the assessment:

- Target website was scanned using OWASP ZAP
 - Automated scanning was performed
 - Results were analyzed for vulnerabilities
 - Findings were categorized based on severity
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5. Tools Used

- OWASP ZAP (Passive & Active Scanning)
 - Browser (for testing payloads)
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6. Findings

● 6.1 SQL Injection (High Risk)

Description:

SQL Injection allows attackers to manipulate database queries by injecting malicious input. This can lead to unauthorized access to sensitive data.

Evidence:

Payload used: ' OR '1'='1

Impact:

- Unauthorized data access
- Data leakage
- Database manipulation

Risk Level: High

Remediation:

- Use prepared statements (Parameterized Queries)
 - Validate and sanitize user input
 - Implement proper error handling
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● 6.2 Cross-Site Scripting (XSS) (High Risk)

Description:

XSS allows attackers to inject malicious scripts into web pages, which execute in the victim's browser.

Evidence:

Payload used: <script>alert(1)</script>

Impact:

- Session hijacking
- Data theft
- Malicious redirection

Risk Level: High

Remediation:

- Validate user input
- Encode output data

- Use Content Security Policy (CSP)
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□ 6.3 Other Observations (Medium / Low)

- Missing security headers
- Information disclosure issues
- Cookie security attributes not properly set

Remediation:

- Enable security headers (CSP, X-Frame-Options)
 - Use secure cookies (HttpOnly, Secure)
 - Minimize sensitive data exposure
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7. Risk Summary

Vulnerability	Risk Level
SQL Injection	High
Cross-Site Scripting (XSS)	High
Security Misconfigurations	Medium
Informational Issues	Low

8. Conclusion

The vulnerability assessment revealed critical security issues in the application, including SQL Injection and Cross-Site Scripting. These vulnerabilities pose significant risks to data confidentiality and system integrity. Immediate remediation is recommended to enhance the security posture of the application.

9. References

- OWASP Top 10
 - OWASP ZAP Documentation
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10. Appendix

(Screenshots of scan results, alerts, and reports should be attached here)

